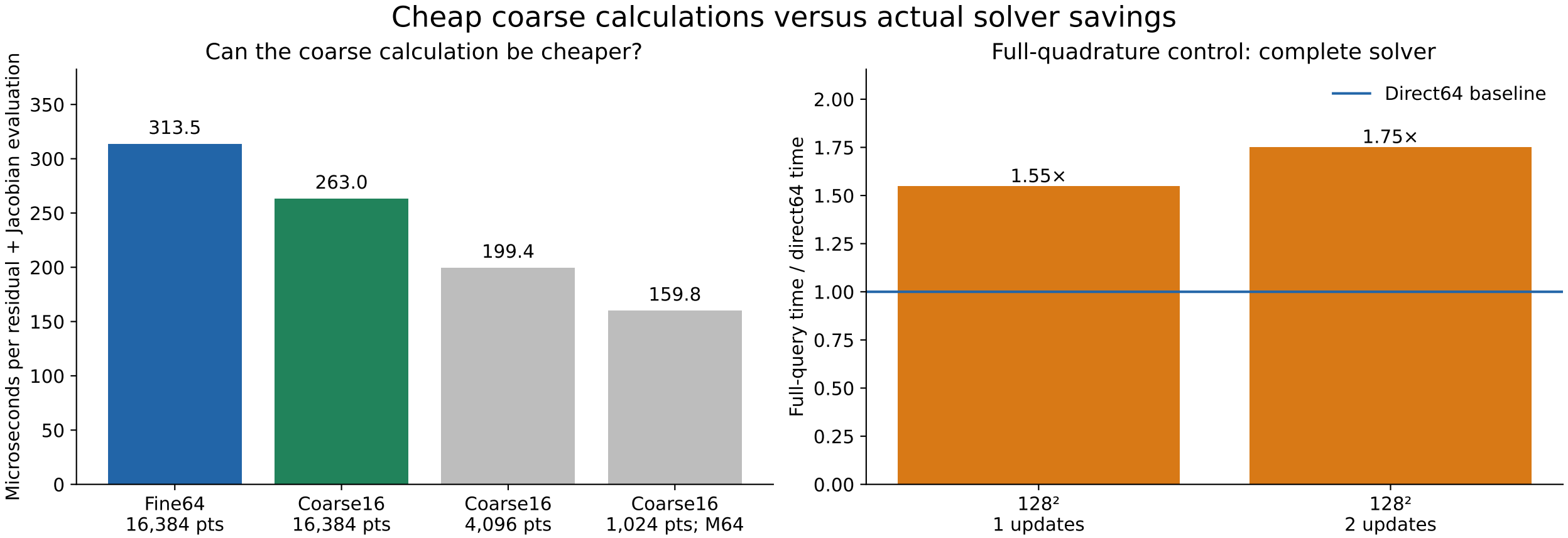
Cheap coarse calculations versus actual solver savings
Can the coarse calculation be cheaper?
Microseconds per residual + Jacobian evaluation
0
50
100
150
200
250
300
350
313.5
263.0
199.4
159.8
Fine64
16,384 pts
Coarse16
16,384 pts
Coarse16
4,096 pts
Coarse16
1,024 pts; M64
Full-quadrature control: complete solver
Full-query time / direct64 time
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Direct64 baseline
1.55×
1.75×
128²
1 updates
128²
2 updates
Job 6647637 · left: isolated synchronized calls at case1 initial state (21 repetitions)
Right: full-quadrature proposals only; 6 cases × 7 paired repetitions.
Grey bars: quadrature panel gate failed; these settings were not tested end-to-end.